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Antonio Jesús Chaves García

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Gender: Male **Date of birth:** 31/10/1999 **Nationality:** Spanish

ABOUT ME

Antonio Jesús Chaves received his B.Sc. in Computer Science Engineering and his M.Sc. in Software Engineering and Artificial Intelligence from the University of Málaga, Spain, in 2021 and 2022, respectively. In January 2024, he was awarded an FPU grant from the Spanish government, which allowed him to begin teaching activities. He is currently a Ph.D. student at the ERTIS Research Group, University of Málaga, and a member of the ITIS Software Institute, University of Málaga. He has completed a three-month research secondment at Yonsei University, South Korea. His research interests include modern deep-learning techniques, such as object recognition, fault detection, federated learning, and artificial intelligence applied to the IoT field.

WORK EXPERIENCE

[10/2022 – Current]

PhD Student

Universidad de Málaga - ERTIS Research Group

City: Málaga | **Country:** Spain

PhD student in Computer Technologies (Security, Networks and Embedded Systems research topic) at the University of Málaga.

Project aimed at the study, application and detection of potential problems on the use of Machine Learning techniques on real time data streams in Industry 4.0 use cases.

[04/2021 – 10/2022]

Research Technician

Universidad de Málaga - ERTIS Research Group

City: Málaga | **Country:** Spain

- Embedded software development.
- Analysis of biological samples using Computer Vision libraries and techniques.
- Sensor deployment for data collection.
- Extended "Kafka-ML" application support to new Deep Learning frameworks.
- Data analysis and ML models development to predict hard-to-measure variables in natural environments and Industry 4.0

[10/2020 – 04/2021]

Research intern

Universidad de Málaga - ERTIS Research Group

City: Málaga | **Country:** Spain

- Embedded Software development using C++ and OpenCV.

EDUCATION AND TRAINING

[10/2022 – Current]

PhD in Computer Technologies

Universidad de Málaga

City: Málaga | **Country:** Spain |

[09/2021 – 10/2022] **Master's degree in Software Engineering and Artificial Intelligence**

Universidad de Málaga

City: Málaga | **Country:** Spain |

[10/2017 – 07/2021] **Bachelor's degree in Computer Science and Engineering**

Universidad de Málaga

City: Málaga | **Country:** Spain |

PUBLICATIONS

[2025] **A soft sensor open-source methodology for inexpensive monitoring of water quality: A case study of NO3- concentrations**

Reference: Chaves, A. J., et al. (2025). A soft sensor open-source methodology for inexpensive monitoring of water quality: A case study of NO3-concentrations. Journal of Computational Science, 102522.

[2024] **Federated Learning Meets Blockchain: A Kafka-ML Integration for reliable model training using data streams**

Reference: Chaves, A. J., et al. (2024, December). Federated Learning Meets Blockchain: A Kafka-ML Integration for reliable model training using data streams. In 2024 IEEE International Conference on Big Data (BigData) (pp. 7677-7686). IEEE.

[2024] **Towards flexible data stream collaboration: Federated Learning in Kafka-ML**

Reference: Chaves, A. J., Martín, C., & Díaz, M. (2024). Towards flexible data stream collaboration: Federated Learning in Kafka-ML. Internet of Things, 25, 101036.

[2024] **The orchestration of Machine Learning frameworks with data streams and GPU acceleration in Kafka-ML: A deep-learning performance comparative**

Reference: Chaves, A. J., Martín, C., & Díaz, M. (2024). The orchestration of Machine Learning frameworks with data streams and GPU acceleration in Kafka-ML: A deep-learning performance comparative. Expert Systems, 41(2), e13287.

[2023] **Pollen recognition through an open-source web-based system: automated particle counting for aerobiological analysis**

Reference: Chaves, A. J., et al. (2024). Pollen recognition through an open-source web-based system: automated particle counting for aerobiological analysis. Earth Science Informatics, 17(1), 699-710.

[2023] **A Methodology for the Development of Soft Sensors with Kafka-ML**

Reference: Chaves, A. J., Martín, C., Torres, L. L., Soler, E., & Díaz, M. (2023). A methodology for the development of soft sensors with Kafka-ML. In Data Analytics for Internet of Things Infrastructure (pp. 307-324). Cham: Springer Nature Switzerland.

CONFERENCES AND SEMINARS

[15/12/2024 – 18/12/2024] **2024 IEEE International Conference on Big Data** Washington, D.C., United States

[24/01/2023 – 26/01/2023] **XXII Jornadas de Tiempo Real** Madrid, Spain

PROJECTS

[01/01/2023 – 31/12/2026] **Electric Vehicles Point Location Optimisation via Vehicular Communications, Horizon Europe**

[01/01/2022 – 31/12/2024] **5G+TACTILE_2: Digital vertical twins for B5G/6G networks, UNICO-5G I+D call, Spanish Government**

[01/01/2020 – 31/12/2023] **EnBiC2-Lab, LifeWatch ERIC Project**

[01/01/2019 – 31/12/2021] **rFOG: Improving latency and reliability of offloaded computation to the FOG for critical services, Spanish Government**